

CEN 116- EXERCISE

1) Write the output of the following program.

```
#include <stdio.h>

void calculate(int *px, int *py)
{
    int *pz;
    static int k = 0;

    for (pz = py; pz < py + 4; pz++)
    {
        *pz = *(py+k) + *(px+k);
        k++;
    }
}

int main()
{
    int a[4][4]={0,1,2,3},{4,5,6,7},{8,9,10,11},{12,13,14,15}};
    int (*px)[4], *py[8];

    py[0] = a[3];
    py[1] = a[0];

    px = &a[3];

    calculate(px[0],py[0]);
    printf("%d %d %d\n",*py[0],*py[1],(*py)[3]);

    return 0;
}
```

2) Write the output of the following program.

```
#include <stdio.h>

int func (int k, int *px)
{
    *px = *px & k ^ 10;

    return *px;
}

int main()
{
    int a = 4, b = 8, c,d;

    c = a >> 2;
    c += b;
    c = c << 4;

    d = a >> 2 + (a << 4) & 10 ^ func(4,&d) | 4;

    printf("%d--%d\n",c,d);

    return 0;
}
```

- 3) Write a C program to find the unique names in a file. Your program will create a binary file to keep ids, names and salaries of employees. You will use structures to write each record to the file. After creation of the file you will find unique names in that file. You will print only names.

EX: 123 John 1500
234 Jersey 2300
345 John 1200
290 George 1890
190 John 1700 you will display : John Jersey George

- 4) Write a C program including a macro to PRINT variable name and value when it is called. You will use that macro to debug your code. You have to control whether you are using it in DEBUG mode or not.
- 5) Write a C program to keep people and company information in a linked list. (You will use unions)
- 6) Write a C program to print entered string in reverse order. Your program must include **recursive function** and **char pointer**.
- 7) Write a C program to change the id values of the repeated names to 333 in file created for QUE 3.
- 8) Write 2 header files containing the same function declarations with different operations and write a C program and try to include these header files and if you get errors, try to solve it by using MACROS.